

Continuous Water-Quality Monitoring with a Low-Cost Autonomous Sensor Buoy

A general, transferable method for high-frequency in-situ monitoring of temperature, salinity, dissolved oxygen and water level at an aquaculture site, using an autonomous logging buoy that non-specialist staff can run.

Protocol ID: schmidt_2018

Sector: Mollusc · Usability Tier 4 (Practitioner)

MEL indicator: WQ 1.2.1 (Dissolved oxygen)

Source paper: Design and operation of a low-cost and compact autonomous buoy system for use in coastal aquaculture and water quality monitoring

Schmidt, W., et al. (2018). Aquacultural Engineering 80: 28–36. DOI: <https://doi.org/10.1016/j.aquaeng.2017.12.002>

Introduction

This protocol sets up continuous, high-frequency monitoring of the physical and chemical water quality at an aquaculture site using an autonomous sensor buoy. A small suite of self-logging sensors — temperature at several depths, salinity, dissolved oxygen, and water level — records around the clock for months at a time, then is periodically visited to offload data and clean fouling. The result is a continuous time series that captures the short-term variation (diurnal, tidal, storm-driven) that sparse spot-sampling misses.

The approach is deliberately gear-agnostic and low-cost: choose logging sensors and mooring hardware that suit your site and budget, and design the mooring so the instrument string can be lifted and serviced by ordinary farm staff. Continuous dissolved-oxygen and temperature/salinity records help characterise a site's suitability for shellfish or finfish and reveal biological activity such as stratification and phytoplankton-driven oxygen swings.

Measured parameters

- **Dissolved oxygen (mg L^{-1})** — the mapped MEL indicator (WQ 1.2.1). Also a signal of biological activity / eutrophication (super- and under-saturation).
- **Water temperature ($^{\circ}\text{C}$)** — logged at several depths to detect stratification / mixing.
- **Salinity (PSU)** — derived from a conductivity sensor.
- **Water level / depth (m)** — from a pressure sensor; also confirms the instrument string's actual depth and vertical orientation.
- **Optional add-on sensors** — the same string can carry extra sensors [*e.g. a pCO_2 sensor for the carbonate system*].

Requirements

Personnel: 1–2 people; the system is designed so routine operation and maintenance need no specialist. [*estimate*]

Estimated time:

- Build + mooring assembly + sensor configuration and calibration: on the order of a day or two.
- Maintenance visits: short — around 20 minutes on site to lift the instrument string, offload data, clean fouling and return it. Revisit periodically [*e.g. monthly*].
- Deployment duration: continuous over the monitoring period; a modest logging interval [*e.g. 10 min*] lets self-logging sensors run for up to ~12 months between battery service.

Sampling frequency & timing

Framework guidance. The MEL framework sets a recommended sampling frequency for each indicator, on the principle that sampling should be spaced “at intervals sufficient to build an accurate depiction of the process over time.” Its values run Once · Seasonal · Biannual · Annual · Ongoing.

What the source did. Logged continuously (e.g. every 10 minutes), serviced monthly, across multi-month deployments.

Recommended for sound science. Dissolved oxygen, temperature and salinity change on diurnal, tidal and storm timescales, so log continuously (Ongoing) rather than spot-sampling — sparse sampling misses the variation that matters. Cover a full annual cycle so every season is represented, and continue across years [*set to your monitoring goal*] to establish the site’s normal range and detect change against year-to-year variability.

Materials

- A suite of self-logging water-quality sensors [*use preferred loggers — e.g. HOBO U-series*]: temperature (several, at different depths), conductivity/salinity, pressure/level, and dissolved oxygen. Each with an internal logger and battery.
- A data offload device [*e.g. a waterproof data shuttle*] so data can be pulled in the field without a computer or cables.
- Antifouling guards / housing for the fouling-prone sensors (dissolved oxygen and conductivity).
- A robust two-part mooring [*use preferred, locally-proven mooring hardware*] — a main float + anchor, plus a separate surface buoy carrying the instrument string, weighted at the bottom to hang vertical.
- A calibrated independent reference instrument for validation [*e.g. a handheld CTD*].
- Dissolved-oxygen calibration supplies (a water-saturated-air step and a zero-oxygen solution); a cleaning kit (soft brush + fresh water); a waterproof ID tag with a contact number.

Methods

Preparation & calibration

1. Choose the parameters and the depths to monitor [*record the actual sensor depths*]; stratification detection needs temperature at more than one depth.
2. Configure the loggers: set a common clock and a fixed logging interval [*e.g. 10 min*].
3. Calibrate the dissolved-oxygen sensor per its manufacturer — typically a 3-step routine: 100% saturation in water-saturated air, a temperature-equilibration step, then 0% saturation in a zero-oxygen solution. Factory-calibrated sensors can deploy as supplied; note any that need periodic recalibration.
4. Fit antifouling guards/housing to the dissolved-oxygen and conductivity sensors.

Mooring & deployment

1. Build a robust two-part mooring so the lightweight instrument string can be lifted and serviced independently of the main mooring (which stays in place and can even moor the service vessel).
2. Weight the bottom of the instrument string so it hangs vertical, and include a pressure/level sensor — the string can circulate with the current, so the pressure record is how you know its true depth and orientation.
3. Attach a waterproof ID tag with a contact number. Deploy near the site and record deployment metadata (site, coordinates, date, sensor IDs and depths, logging interval, mooring configuration).

Operation & maintenance

1. Let the sensors log continuously. Visit periodically [*e.g. monthly*] to offload data with the shuttle, clean biofouling from the sensors (a soft brush + fresh water), and check battery status.
2. At each maintenance visit, take an independent reference reading with a calibrated instrument (e.g. a CTD) at the sensor depths — these paired readings are what you use to validate sensor accuracy and drift. Verify the reference instrument itself is sound.

Data processing & quality control

1. Export the logged data (e.g. to CSV). Convert conductivity to salinity using standard seawater equations [*use preferred tool — e.g. the marelac R package*], and derive water level from the pressure sensor.
2. Apply quality-control rules [*follow a recognised QC standard, e.g. IOOS QARTOD*]: (a) discard readings outside each sensor's valid range; (b) discard the first ~2 hours after deployment and after each cleaning while sensors restabilise; (c) drop single anomalous spikes (abrupt jumps between consecutive readings); (d) where a gradual roll-off appears before a cleaning, keep the pre-drift peak and discard the fouled tail; (e) discard any period when the pressure sensor shows the string has floated to the surface (tangled).

Validation & analysis

Compare the sensor readings against the independent reference to compute **RMSE and bias** per parameter — handled in the paired stats workbook. The continuous time series itself is large; plot it (daily mean with min/max envelope) in a dashboard or script rather than storing it in the workbook — the workbook's *Time-series & dashboard* sheet gives an R recipe. Read the plots for stratification (divergence between depths) and for dissolved-oxygen super-/under-saturation as a signal of phytoplankton activity.

Expect these failure modes (and manage them)

- Biofouling is the single biggest limiter of accuracy and deployment length — hence antifouling guards, cleaning at every visit, and QC rules (c)–(d) to catch fouling drift. Expect to discard a non-trivial fraction of the most fouling-prone channel (e.g. conductivity).
- Storms and mooring stress — the instrument string can tangle around the main float (the pressure sensor reveals this) or, in severe storms, the surface buoy can part from the mooring. Use a robust mooring, keep spliced rope clear of split pins/shackles, and let the pressure record flag surface-tangle periods for removal.
- Reference-instrument failure — the validation is only as good as the reference; confirm the reference instrument is calibrated and working before trusting drift estimates.
- Sensor drift — some sensors (dissolved oxygen, conductivity) drift and need periodic calibration; schedule it and note calibration dates.
- Instrument circulation — a freely-circulating string compromises vertical orientation; the pressure/level sensor is what makes the depth of every reading known.

How this serves the MEL framework

- **WQ 1.2.1 — Dissolved oxygen:** direct — a dedicated dissolved-oxygen sensor logging continuously.
- *Temperature, salinity, water level* are supporting / context parameters — they characterise the site and let you interpret dissolved oxygen and stratification, but they do not each map to a distinct MEL water-quality indicator.

Data handling

- Keep the raw CSV exports labelled by deployment / site / offload date; retain the QC log and the paired reference readings as the validation record.
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Source provenance & residual gaps

Normalized by AquaPIP from Schmidt et al. (2018). Specific sensor models, mooring hardware, sensor depths, and the logging interval are site/budget choices, shown in [brackets]. Quality-control thresholds follow each sensor's valid range plus a recognised QC standard (e.g. IOOS QARTOD).

Note: this protocol omits the WQ 2.1.2 (Microorganisms) mapping carried on the database row. The buoy measures physico-chemical proxies (dissolved oxygen, temperature, stratification) and lets a user infer harmful-algae risk; it does not directly monitor microorganisms — in the source, algal presence was confirmed by separate agency sampling, not the buoy. Protocol ID schmidt_2018.